

Week 7: Six-Axis Benchmark

Yale EMMLC triage extract · target ESI 1 TO 5 · 80/20 stratified split, `random_state = 42`
44,096 training patients · 11,025 test patients · 208 numerical features

Headline: all models, seven axes

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	Recall (ESI 1)	Train time (s)	Infer time (ms/pred)	Interpretability
Dummy (stratified)	0.375	0.204	0.204	0.204	0.000	0.00	0.0004	N/A
Logistic Regression	0.667	0.582	0.463	0.492	0.250	8.79	0.0041	High: read the weights
Decision Tree (d=5)	0.556	0.265	0.245	0.216	0.000	0.32	0.0006	High: trace the branches
Random Forest	0.491	0.403	0.553	0.390	0.500	15.18	0.0570	Medium: feature importances
Gradient Boosting	0.668	0.496	0.435	0.442	0.125	20.65	0.0708	Low: needs SHAP per patient

Recall (ESI 1) is the clinically critical axis. Interpretability is the qualitative seventh axis (the one-minute test: can a single prediction be explained to a clinician, and by what method?).

In patient terms. Of the 16 critical (ESI-1) arrivals in the test set, the Random Forest identified 8, logistic regression 4, gradient boosting 2, and the decision tree none.

Recommended model, per ESI level

Random Forest, broken out across all five acuity levels (test set, 11,025 patients):

ESI level	Precision	Recall	F1	Patients
1 - critical	0.13	0.50	0.21	16
2 - emergent	0.63	0.64	0.63	3,585
3 - urgent	0.80	0.30	0.44	5,402
4 - less urgent	0.31	0.77	0.45	1,779
5 - non-urgent	0.15	0.56	0.23	243
Macro avg	0.40	0.55	0.39	11,025

The model is biased towards sensitivity at the urgent end (ESI-1 recall 0.50, ESI-4 recall 0.77) at the cost of precision (ESI-1 precision 0.13) the balanced-weighting trade, chosen because a false alarm is cheaper than a missed emergency. Weakest recall is ESI-3 (0.30).

Top drivers (Random Forest feature importances)

Rank	Feature	Importance
1	Stroke-alert flag	0.113
2	Systolic blood pressure	0.087
3	Oxygen-delivery device	0.076
4	Diastolic blood pressure	0.072
5	Abdominal-pain flag	0.065
6	Respiratory rate	0.055
7	Heart rate	0.050
8	Oxygen saturation	0.049
9	Glucose	0.047
10	Temperature	0.046

Single-patient SHAP explanation produced in 0.2 s (worked example: an ESI-2 rating driven by shortness of breath, oxygen-delivery device and oxygen saturation).